

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	1.02%	0.97%	13.57%	1.65%
Meat MCal Target	55,748.64	53,017.71	742,718.79	90,214.86
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	877.83	322.05	989.60	1,420.53
MCals from meat	55748.64	53017.71	742718.79	90214.86
Max Utilized Mcals allowed from Dairy	58173.46	0.00	502319.57	90132.91
MCals actually Utilized from Dairy	58173.46	0.00	502319.57	90132.91
Percent of diet from their meat	1.02%	0.97%	13.57%	1.65%
Percent of Diet from their Dairy	1.06%	0.00%	9.17%	1.65%
Kg Fleece	6,634.32	0.00	0.00	0.00
1 Year+ (age) skins	326.47	12.54	190.14	528.12
Yearly Offspring skins	1,597.61	379.99	1,457.70	2,584.96
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	1,567.54	58.82	1,455.29	2,536.67
Males Step 1	438.91	161.02	494.80	710.27
Females Step 1	438.91	161.02	494.80	710.27
Replacement Mothers per year	313.81	12.06	182.10	507.63
Breeding Males Required	82.30	3.09	76.40	133.18
Additional Mothers	1,285.34	10.58	688.39	2,079.33
Males Step 2	359.89	28.97	234.05	582.21
Females 2	359.89	28.97	234.05	582.21
Total Mothers	1,881.35	70.89	1,637.39	3,044.30
Total Males	798.81	190.00	728.85	1,292.48
Total Females	79.02	132.05	260.75	128.06
Total Offspring	1,597.61	379.99	1,457.70	2,584.96
Total Animals	2,841.47	396.02	2,703.39	4,598.01
Units per Hectare	1.00			
Total Adults per Hectare	1.79	0.27	0.44	1.79
Total Offspring per Hectare	0.88	1.21	0.27	0.88
	0.70%		8.52%	1.13%
Total Adults	3,249	85	2,402	5,257
Total Offspring	1,598	380	1,458	2,585
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	283.74	0.00	179.69	459.34
Other	25,561.83	0.00	0.00	41,359.33
MCals Dry Mothers	331,407.26	0.00	380,396.02	536,514.19
total Maintenance MCals	356,969.09	0.00	380,396.02	577,873.51

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	951,379	173,486	2,440,194	1,539,342
Lactating	1,815,050	61,673	2,623,865	2,936,771
Maintenance	356,969	0	380,396	577,874
Adult Males	96,122	0	161,745	155,549
Offspring	438,545	381,893	5,335,191	709,571
TOTAL	3,658,065.3	617,050.9	10,941,390.3	5,919,106.1

Animal Diet Breakdown					
	Sheep	Pig	Cow	Goat	
	Crop Barley	Barley	Barley	Barley	
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0	
MCal kg ⁻¹					
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0	
MCal kg ⁻¹					
Days	90.0	90.0	90.0	90.0	
Number of Animals	1,597.6	380.0	1,457.7	2,585.0	
MCals from Straw/Chaff	204175.1	0.0	923274.9	330357.6	
MCals from Grain	323516.9	76948.5	590369.5	523454.0	
MCals from other feed	527692.1	76948.5	1513644.5	853811.6	

Diet Calculations		
Plants Mcal	3,729,374.059	
Meat Mcal	1,095,000	
Dairy	650,625.94095287	
	Perfect	
Plants %	68.12%	
Meat %	20.00%	
Dairy %	11.88%	
check	Perfect	
Percent		
Domestic Meat	17.20%	
Dairy	11.88%	
Wild Meat	2.80%	
Crops	61.30%	
Wild Plants	6.81%	
check	Perfect	
new dairy	650625.94	
potential dairy < limit	TRUE	
new limit	80.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	941,700.000	
Dairy	650,625.941	
Wild Meat	153300.000	
Crops	3,356,436.653	
Wild Plants	372,937.406	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd	1881	71	1637	3044				
Males for Slaughter	799	190	729	1292				
Females for Slaughter	79	132	261	128				
Offspring available for Consumption	1598	380	1458	2585				
Gestation Req'd	522745	147031	1656586	845928				
Lactation Req'd	997297	52269	1781275	1613869				
Maintenance Req'd	1186167	0	487467	1919289				
Adult Males	96122	0	161745	155549				
Offspring Required	524816	416092	6384737	849159				
Energy Required	3327147	615392	10471809	5383794				
Ewe Fleece Kg Yr ⁻¹	4233							
Ram Fleece Kg year ⁻¹	165							
Juvenile Fleece Kg Yr ⁻¹	2237							
Kg Fleece Year ⁻¹	6634	0	0	0				